

TTK4147 Real-time Systems

12 - Analysis of Physical Systems

12.11.2025

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Today

- Why analysis of the Physical System?
- Example 1: Medical scanning US imaging, possible specifications?
- Example 2: Spot which tracks actors, possible specifications?
- Example 3: Marble sorting equipment. Deeper analysis
- Considerations about concurrency in the marble sorting equipment

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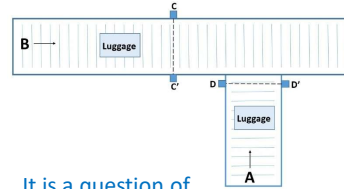
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Why analysis of the Physical System / Environment?

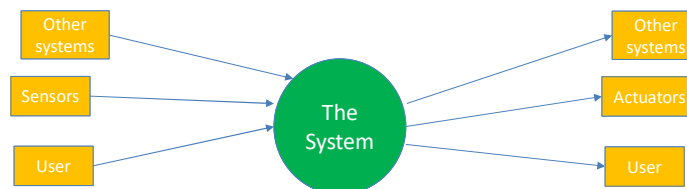
that your technical system shall interact with

It is often a real-time system!



It is a question of synchronization by use of

- Sensor inputs
- Actuator control



Use of Context diagram to describe a system

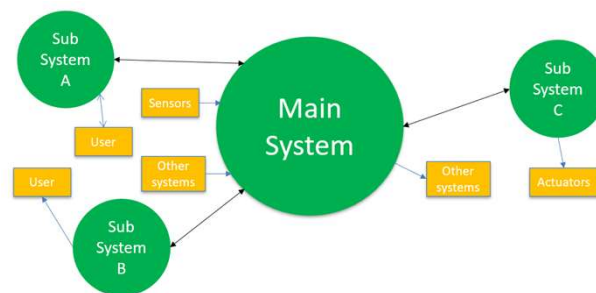
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Why analysis of the Physical System? continue



Distributed system

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Why analysis of the Physical System? continue

- Want to determine the structure and nature of the physical system, to help decide the structure / architecture of the system to be designed.

E.g.

- is the physical system distributed?
- is it a real time system? What kind of RT-system (soft, firm, hard)?
- do different parts of the system need synchronization?
- what are the physical laws concerning the system?
- where are the bottle necks?
- has any part large divergence in CPU time consumption?
- is the system event driven or should the system be periodically monitored?
 - should a change in sensor perception trig an interrupt or should the change be detected by periodically sampling?
 - are the events frequently (regular) or sporadically?
- how could the system be modularized
- does the system need any (power / communication) infrastructure, and is that available?

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Why analysis of the Physical System? continue

- Is it possible to design a system based on first version (preferable / “dreamed”) specifications?

(E.g. portability, low weight, long operation time, small size, high power consumption, low price .)



If not

Analyze trade-offs in specification.

- Specification for the system to be designed should always be worked out.
 - what are the real time requirements?
 - What are the dependency in the system?

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From logical to physical system

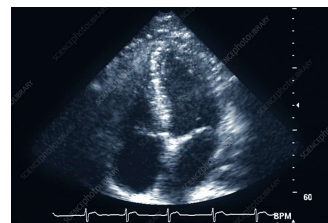
Example 1: We want to make a medical scanning ultrasound imaging system for tissue imaging.

Development of specifications



Fetus

Heart



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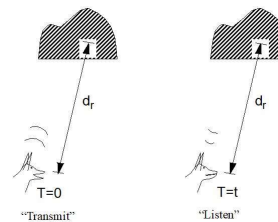
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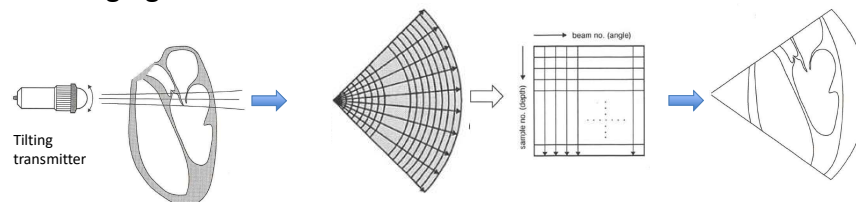
Medical scanning US imaging, principle

Physical principle: Echo imaging, measure time of flight to tissue borders (d_r)

$$d_r = vt/2$$



Imaging:



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Medical scanning US imaging, definitely a R-T system

We have to consider:

- What are the physical laws concerning the system?
- Is it possible to design a system based on first version (preferable / “dreamed”) specifications?
If not
Analyze trade-offs in requirements.
- Specification for the system to be designed should always be worked out.
 - What are the real time requirements?
 - What are the dependency in the system?

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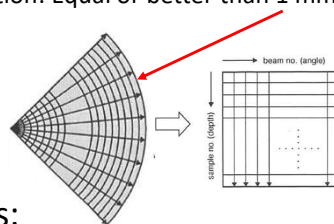
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Medical scanning US imaging , specifications, v1.0

- Single beam
- Mechanical scanned
- Frame rate: 25 fps
- Scanning depth: up to 30 cm
- Scanning angle: up to 45 deg
- Resolution: Equal or better than 1 mm square



Analysis:

- Radial: $300\text{mm} / 512 = 0,59\text{ mm}$
- Angular:
 - 1 mm at 300 mm: $\arctan(1/300) = 0,19^\circ \rightarrow 45^\circ / 0,19^\circ = 236\text{ beams}$
 - 1 mm at 100 mm: $\arctan(1/100) = 0,57^\circ \rightarrow 45^\circ / 0,57^\circ = 79\text{ beams}$

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Medical scanning US imaging, possible specifications?

Speed of sound (v) in muscle: 1589 m/sec, in fat 1450 m/sec. Use 1500 m/sec



Radial:

512 samples on 0,3 m $\rightarrow 512 / 0,3 \text{ samples/m} * 1500 \text{ m/sec} = 2,56 \text{ Msamples/sec}$

Angular:

Uses some time to reposition the transducer and time to allow sound to die out: K'

Min time for one beam: $(d^2)/v + K' = (0,3\text{m} * 2) / 1500\text{m/s} + K' = (0,4 + K) \text{ ms}$

Min time for one frame (236 beams): $(0,4 + K)\text{ms} * 236 = (94 + 236K) \text{ ms}$

If $K = 0 \rightarrow 1 / 0,094\text{sec} = 10,6 \text{ fps} \leq 25 \text{ fps}$



Min time for one frame (79 beams): $(0,4 + K)\text{ms} * 79 = (32 + 79K) \text{ ms}$

If $K = 0 \rightarrow 1 / 0,032\text{sec} = 32 \text{ fps}$



32 fps > 25 fps $\rightarrow$ allows for $K > 0$

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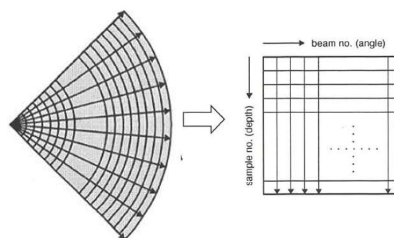
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Medical scanning US imaging , new specifications

- Single beam
- Mechanical scanned
- Frame rate: 25 fps
- Scanning depth: up to 30 cm
- Scanning angle: up to 45 deg
- Resolution: Equal or better than 1 mm square in a depth of up to 10 cm
 $\rightarrow$ angular resolution is 0,57 deg.



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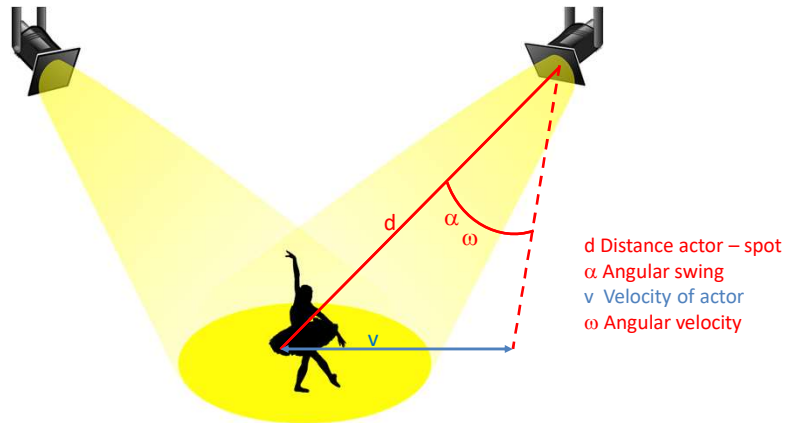
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Example 2: Spot which tracks actors

We want to have spot-lights tracking of the actors all the time.



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Example 2: Spot which tracks actors

We want to have spot-lights tracking of the actors all the time. Is this possible?

- Real time system
- What are the real time requirements/specification?

Requirements:

- Actor should always be within the spot-light
- Actor can move with velocity v
- Actor's position given by external instrumentation
- The moment of inertia of a spot is I
- The power of each motor on the spot is P

Mechanical:

$I_{\max}$ given by $\omega_{\max}^2 I$, $\omega_{\max} v$ $P = f(I)$

$v_{\text{possible to follow}} = g(d, \omega_{\max}, \omega_{\max} v)$

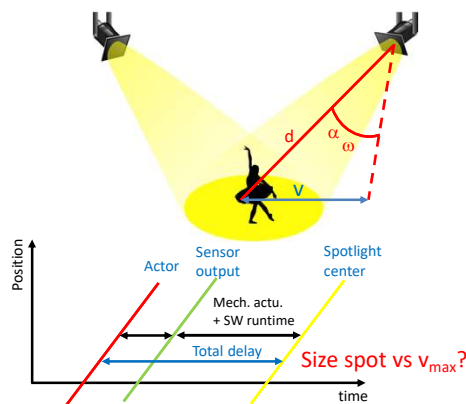
As d decreases, α and ω must increase to be able to follow an actor moving with v

Given $v_{\max}$

Is $d_{\min}$ able to track an actor moving with velocity v acceptable, with respect to available P and given I ?

Strategy:

Keep the spot center at the actor



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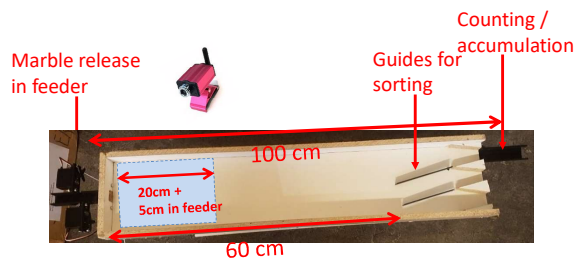
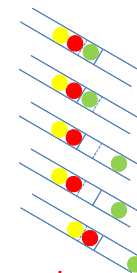
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Example 3: Marbles sorting equipment

We want to make an equipment to sort marbles according to colors.
We will use marbles of three colors.



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Example 3: Actuators and sensors



Sensors:

- 1 camera
- 3 object (marble) counting detectors

Actuators:

- 2 guide control motors
- 2 marble release motors

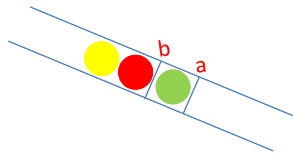
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Example 3: Story



Initial: MarbleReleaseGates (MRG) a and b is closed, variables MarbleCount (MC)_{G, R, Y} are 0. The variable Marble-inCameraPosition (MCP) is set to 200 [mm].

1. MRGa opens for x sec, then MRGa closes and MRGb opens for x sec, then MRGb closes.
2. The camera takes continuously pictures (video). The analysis of a picture is done on request.
If a marble is detected with a position $P < MCP$, a new marble is detected. MCP is set to P. If new marble, the color is determined and variable C (Color) is set according to determination.
3. If variable Marble Guides (MG) is different from C, $MG = C$ and the guides are set according to C
4. When one of the MarbleDetectors (MD) detects one object, the corresponding MC is incremented

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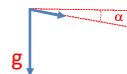
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Real time calculations

How fast does a marble roll?

How long does a marble take to reach the guides?

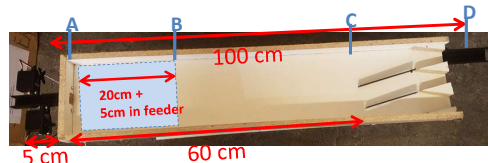
The physical laws set the time requirements to the system



$$a_p = g \sin(\alpha)$$

$$v = \sqrt{2a_p d}$$

$$t = \sqrt{2d/a_p}$$



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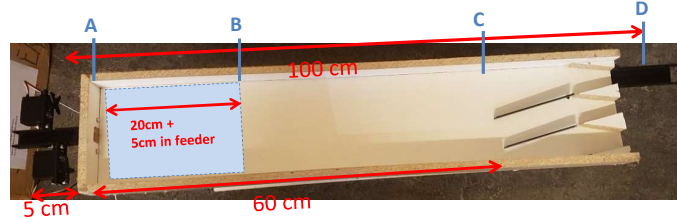
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Results real time calculations

	[deg]	[sec]	A 5 cm	B 25 cm	C 65 cm	D 100 cm	A-B t 5 - 25 cm	A-C t 5 - 65 cm	B-C t 25 - 65 cm	C-D t 65 - 100 cm
$\alpha = 10$		$t =$	0,24	0,54	0,87	1,08	0,30	0,63	0,33	0,21
$\alpha = 25$		$t =$	0,16	0,35	0,56	0,69	0,19	0,40	0,21	0,13
$\alpha = 45$		$t =$	0,12	0,27	0,43	0,54	0,15	0,31	0,16	0,10
$\alpha = 60$		$t =$	0,11	0,24	0,39	0,49	0,13	0,28	0,15	0,09
$\alpha = 80$		$t =$	0,10	0,23	0,37	0,46	0,13	0,27	0,14	0,09



Time for setting the guides (time of movement) (GuideSet): 0,1 sec

→ Time A-C' = time (A-C - 0,1 sec). (Time from A to when start moving the guides)

Due to the collision between the marble and the guides, the period in the sorting area (65 - 100 cm, C - D) will increase

Have to argue for the time x in MarbleRelease

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Marble sorter, simple version

One marble in action a time

Simplest version, sequential handling according to the story, active waiting.



Questions for this simple version:

- How could the SW be organized and structured?



- What is the largest uncertainty (with respect to time) in the set-up?



Probably the camera:

- When within $1/FR$ is the picture taken?
- How is the frames synchronized with the marble release?

- How to increase the efficiency (the number of marbles to be sorted pr time unit)?

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Marble sorter, SW organization: "Big While Loop"

1. MRGa opens for x sec, MRGa closes and MRGb opens for x sec, then MRGb closes.
2. The camera takes continuously pictures and analysis the pictures.
If a marble is detected with a position $P < MCP$, a new marble is detected. MCP is set to P. If new marble, the color is determined and variable C is set according to determination.
3. If Marble Guides (MG) is different from C: $MG \neq C$ and the guides are set according to C
4. When one of the Marble Detectors (MD) detects one object (generates an IRQ), the corresponding MC is incremented and NoIRQ is set to 1 (within the ISR)

Can run as a "Big While Loop"

Start: Initialize variables

```
while ( 1 ) {
    MarbleRelease();
    while (NewMarble==0) Camera();
    if (MG <> C) GuideSet( C);
    while (NoIRQ==1); //Waiting for an IRQ from MD
    NoIRQ = 1; NewMarble = 0;
}
```

1. MarbleRelease()
2. Camera() → Sets NewMarble, C
3. GuideSet(C)
4. Each MD is connected to an HW Interrupt / Interrupt Service Routine (ISR), increasing $MC_{G,R,Y}$ and setting NoIRQ

Camera() consists of FrameAcquisition() and FrameAnalysis() → sets C and NewMarble==1

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Comments concerning "Big While Loop"

The "Big While Loop"

Start: Initialize variables

```
while ( 1 ) {
    MarbleRelease();
    while (NewMarble==0) Camera();
    if (MG <> C) GuideSet( C);
    while (NoIRQ==1); //Waiting for an IRQ from MD
    NoIRQ = 1; NewMarble = 0;
}
```

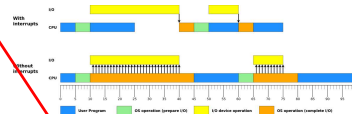
Running with no operation system

Synchronization between SW and "Real World" by use of active waiting



What is this?

Alternative for using IRQ?



Directly polling of Marble Detectors

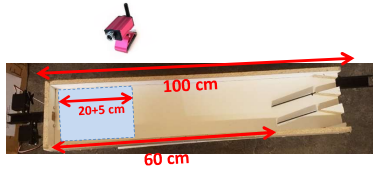
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Sensor data acquisition time



Webcam 720p 30 fps:

- Resolution: 200 mm / 720p = 0,28 mm/p
- Time between frames: $1 / FR = 1 / 30 = 0,033 \text{ s}$
(When the picture arrives, the content (top of picture) may be $1 / FR = 0,033 \text{ sec}$ old)

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Camera routine

The raw camera data arrives as a continuous stream of data, pixel for pixel, frame (picture) for frame.

When FrameAcquisition shall acquire a new frame, it has to wait for a new start of frame (may take up to a frame) and then acquire a complete frame (together up to 2 frames time).

The acquired frame is sent to FrameAnalysis, to investigate whether the frame contains a new marble, and in case, which color.

If so, a NewMarbleFlag is set, the flag and the color is exported, and the routine terminates.

If no new marble is detected, the FrameAcquisition starts over, succeeded by a new FrameAnalysis.

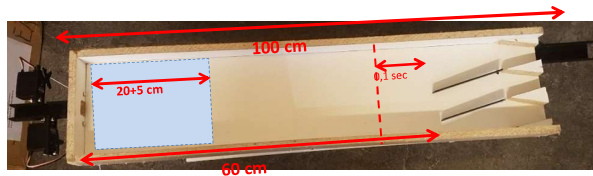
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How to characterize the system?



Hard real time or not?
What is the consequence
not meeting a deadline?

In a real-time system, one or more tasks have real-time constraints. This means that they have to complete within a defined period or point in time.

Hard Real-time

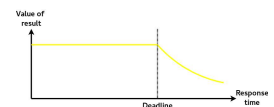
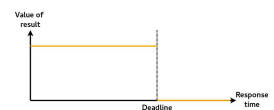
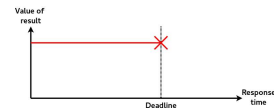
- Failure to complete within the deadline cause total system failure.
- Often associated with “catastrophic failure” (loss of life or large damage).

Firm Real-time

- Failure to complete within the deadline means that the result has no value.
- Repeated missed deadlines will lower the quality of the service provided by the system

Soft Real-time

- Failure to complete within the deadline means that the result is less valuable.



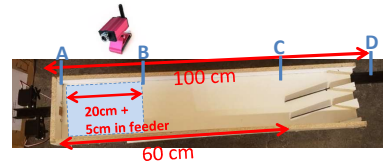
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Worst case reflections



Worst case:

At least one (but worst case 2) frame must be formed + execution of one FrameAnalysis within the time of a marble being in the area bounded by A – B

$$\rightarrow 2 / \text{framerate} + \text{FrameAnalysis} \leq \text{time}(A - B)$$

$$0,066 + \text{FrameAnalysis} \leq \text{time}(A - B)$$

How much time is left for FrameAnalysis? $= (t_{AB} - 0,066)$

		Frame Analysis	A - B
	[deg]	[sec]	t 5 - 25 cm
$\alpha =$	10	0,234	0,300
$\alpha =$	25	0,126	0,192
$\alpha =$	45	0,082	0,148
$\alpha =$	60	0,068	0,134
$\alpha =$	80	-0,006	0,126

Given that the color is determined within the marble is at point B, we need 0,1 sec for Guideset in the distance B – C

(FrameAnalysis may also take some time when the marble is in upper part of B-C)

		B-C
	[deg]	t 25 - 65 cm
a =	10	0,33
a =	25	0,21
a =	45	0,16
a =	60	0,15
a =	80	0,14

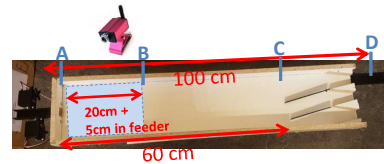
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Worst case reflections



FrameAcquisition synchronized with MarbleRelease:

Does not call FrameAnalysis if no marble

Within the time for a marble to travel A - C

FrameAcquisition + FrameAnalysis + GuideSet have to be executed.

FrameAcquisition: $1 / \text{framerate} = 0,033 \text{ sec}$

GuideSet: 0,1 sec

How much time is left for FrameAnalysis?

$$= (t_{AC} - 0,033 - 0,1)$$

	Frame	Analysis	A - C
	[deg]	[sec]	5 - 65 cm
$\alpha =$	10	0,499	0,632
$\alpha =$	25	0,272	0,405
$\alpha =$	45	0,180	0,313
$\alpha =$	60	0,150	0,283
$\alpha =$	80	0,132	0,265

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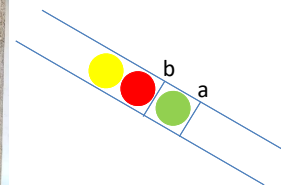
Worst case reflections continue

Decision of x in MarbleRelease:

- MRGa opens for x sec, then MRGa closes and MRGb opens for x sec, then MRGb closes.

x has to be large enough to assure the marble to pass MRGa and for a new marble to pass MRGb and reach MRGa.

x is chosen to be $t(0 - 5 \text{ cm})$:



	[deg]	[sec]	5 cm	25 cm	65 cm	100 cm
$\alpha =$	10	t =	0,24	0,54	0,87	1,08
$\alpha =$	25	t =	0,16	0,35	0,56	0,69
$\alpha =$	45	t =	0,12	0,27	0,43	0,54
$\alpha =$	60	t =	0,11	0,24	0,39	0,49
$\alpha =$	80	t =	0,10	0,23	0,37	0,46

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Marble sorter, SW organization: Task model

We want to convert the SW architecture to a task model with two tasks, **MarbleTask** and **CameraTask**.

The **CameraTask** should wait for a marble to be released and acquires a new, complete picture frame. If the frame contains a new marble, the color shall be determined, the results published and then wait for a new marble release. As long as the frame contains no new marble, the task starts over and acquires a new, complete picture frame.

After the creation, the **MarbleTask** will release one marble and wait for CameraTask to publish the detection of a new marble. When a new marble is detected the GuideSet is called with the appropriate color. GuideSet waits for an ISR-MarbleDetect_{G/R/Y} to increment the associated MarbleCount. After returning from GuideSet, the task public "Ready to start new marble" and exit.

The task model requires a real time operating system

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Marble sorter, SW organization: Task model continue

Pseudo-code of the main loop and the tasks could be:

Main loop

```
Start: Initialize variables
CreateStartTask( CameraTask )
while ( 1 ) {
    CreateStartTask( MarbleTask );
    waitfor( Ready to start new marble );
}
```

CameraTask()

```
Initiate values;
while ( 1 ) {
    waitfor( Marble is released );
    while ( Color == 0 ) {
        FrameAcquisition();
        Color = FrameAnalysis();
    }
    SendMessage( Color );
    Color = 0;
}
```

MarbleTask()

```
Initiate values;
MarbleRelease();
waitfor( Color from CameraTask);
GuideSet( Color );
    // returns when marble is counted
sendMessage( Ready to start new marble);
exit;
```

MarbleRelease()

```
OpenGate(a);
wait(x sec);
sendMessage(Marble is released );
wait(x sec);
CloseGate(a);
OpenGate(b);
wait(x sec);
CloseGate(b);
exit;
```

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Critical regions / Shared resources?

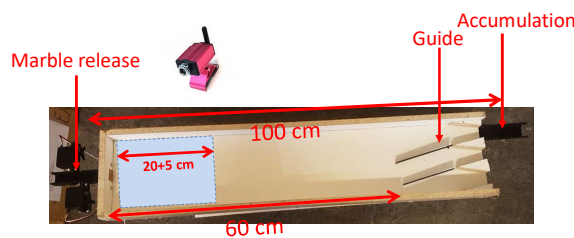
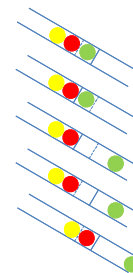
Do we have any critical regions / shared resources in the system?

(What is a critical region?)

Something needed to be guarded



1) Marble release



2) Guide set



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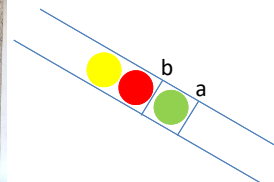
Critical regions / Shared resources

The system has some critical regions / shared resources, and when we are going to use a task model, we should protect the shared resources.

```
MarbleRelease()
    lock_mutex;

    OpenGate(a);
    wait(x sec);
    sendMessage(Marble is released );
    wait(x sec);
    CloseGate(a);
    OpenGate(b);
    wait(x sec);
    CloseGate(b);

    unlock_mutex;
    exit;
```



Critical region

```
function a() {
    lock_mutex
    access_shared_resource
    unlock_mutex
}
```

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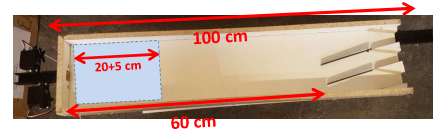
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Critical regions / Shared resources continue

When the marble is in the area related to the guides, only one entity can control the guides.

The time from the routine GuideSet is called (at least 0,1 s before the marble reaches the guides) to the marble is detected by one of the MarbleDetector will be a critical region.



```
GuideSet(Color)
    lock_mutex

    if (Color != OldColor) {
        OldColor = Color;
        set guides according to Color;
    }
    wait( signal from ISR-MarbleDetect)

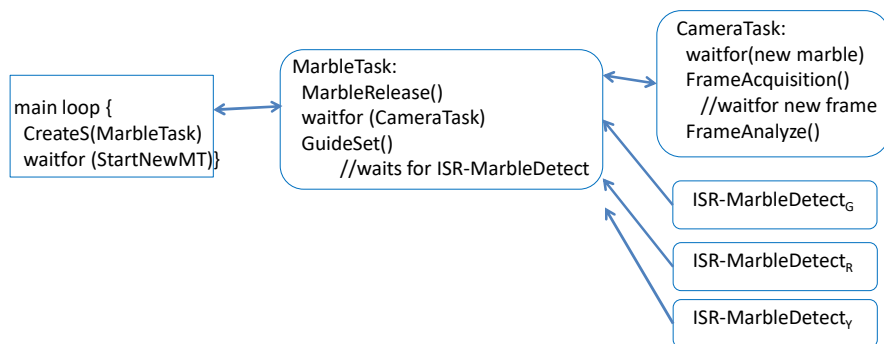
    unlock_mutex
```

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Marble sorter, task version, overview



Still quite serial!

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Optimizing: Smaller change in CameraTask

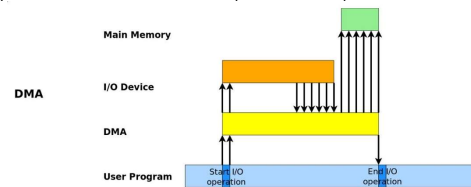
To optimize the CameraTask and to avoid the large worst case FrameAcquisition time, data from the camera is transferred to a ring buffer by use of Direct Memory Access (DMA).

When FrameAcquisition shall acquire a new frame, it picks the last completed frame, which may be almost 1 / FR old (as a new frame may be almost completed).

This reduce the FrameAcquisition to near zero CPU-time (as we don't need to wait for a frame, but the data may still be old!).

Direct Memory Access (DMA)

- Interaction with the I/O device is “outsourced” to a separate DMA module (HW).
- CPU is only informed when the transfer is complete and the requested information is located in memory.



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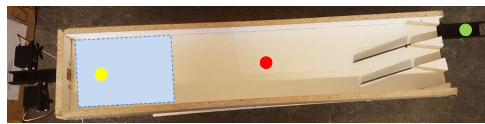
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Concurrent marbles, increasing efficiency

To speed up the sorting, it is desirable to release a new marble before the previous marble is counted.



The task model is prepared for this as critical regions are guarded.

The question is how often a new marble should be released.

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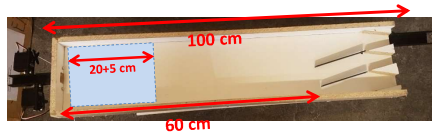
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Concurrent marbles, change in main program

Single marble

Main loop

```
Start: Initialize variables
CreateStartTask( CameraTask )
while ( 1 ) {
    CreateStartTask( MarbleTask );
    waitfor( Ready to start new marble );
}
```



Concurrent marbles

Main loop

```
Start: Initialize variables
CreateStartTask( CameraTask )
while ( 1 ) {
    CreateStartTask( MarbleTask );
    waitfor( Marble is released );
    sleep( f(α) ); //To prevent more marbles
    // in camera view
}
```

MarbleRelease()

```
lock_mutex
OpenGate(a);
Wait(x sec)
sendMessage( CameraTask: Marble is released);
sendMessage(MainLoop: Marble is released);
wait( x sec );
closeGate(a);
openGate(b);
wait( x sec );
closeGate(b);
unlock_mutex
exit
```

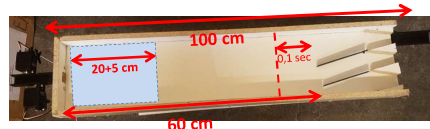
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Concurrent marbles: Considerations



	[deg]	[sec]	5 cm	25 cm	65 cm	100 cm	t 5 - 25 cm	t 5 - 65 cm	t 65 - 100 cm
$\alpha =$	10	t =	0,24	0,54	0,87	1,08	0,30	0,63	0,21
$\alpha =$	25	t =	0,16	0,35	0,56	0,69	0,19	0,40	0,13
$\alpha =$	45	t =	0,12	0,27	0,43	0,54	0,15	0,31	0,10
$\alpha =$	60	t =	0,11	0,24	0,39	0,49	0,13	0,28	0,09
$\alpha =$	80	t =	0,10	0,23	0,37	0,46	0,13	0,27	0,09

- Previously, no “parallel” load on CPU, need new time analysis
- FrameAnalysis will be active more often. More complex to handle more marbles
- Critical regions will delay actions
- The marble inside critical area “Guiders” has to leave this area before next enters

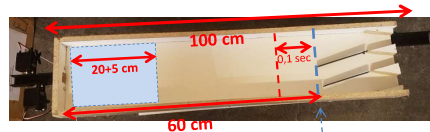
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Concurrent marbles: Considerations continue



- Reduce critical area “Guiders” ?
 - Must be based on calculated velocity
 - Loss of velocity when bumping into guides
 - Introduces uncertainty, must add margins.
 - Alternatives?
 - Detectors at the guides
 - Set guides in middle positions after each count

Detector

Make synchronization possible

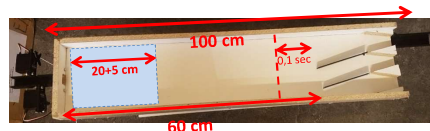
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Concurrent marbles: Considerations continue



- Priority distinction between tasks?
Is this a mixed criticality system?
- What kind of scheduler should be used?
(First-come-first-served, Round-Robin, Shortest-process-next, Shortest-remaining-time, Highest-response-ratio-next, Feedback)
- Is there anything / activity that can give variation / divergence in time?
- What will be the heaviest CPU load?
- Introducing timeouts in wait-for-events (e.g. waitfor camera, waitfor ISR)
- The area covered by the camera / area to be analyzed should be well chosen (not larger then needed).

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Marble sorter, implemented



Optimizing: Puts the guides in middle position,
Analyze a smaller part at the top of the picture, looking for marbles.
Analyze time: From 0,32 to 0,032 sec.

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TTK4147 lecture and exercise plan 2025

Date	Week	Lectures Wednesday 10:15 - 13:00 in H1 - Main Building
2025-08-20	34	01 Introduction
2025-08-27	35	02 Hardware and operation systems
2025-09-03	36	03 Multiprogramming and Memory Management
2025-09-10	37	04 Scheduling
2025-09-17	38	05 Deadlocks, mixed criticality and freertos
2025-09-24	39	06 Priority Inversion
2025-10-01	40	07 Linux
2025-10-08	41	08 More Scheduling
2025-10-15	42	9 Guest lecture
2025-10-22	43	10 QNX
2025-10-29	44	11 Repetition
2025-11-05	45	12 System Architectures
2025-11-12	46	13 Analysis of Physical systems
2025-11-19	47	14 Repetition
2025-11-24	51	Exam 15:00 - 19:00

Date	Week	Exercises Monday 14:00 - 18:00, Friday 14:00 - 18:00, Samtidslabben, G203/G204
2025-08-25 / 2025-08-22	34	
2025-09-01 / 2025-09-05	35	0 - Basics and setup
2025-09-08 / 2025-09-12	36	1 - Time
2025-09-15 / 2025-09-19	37	2 - Memory
2025-09-22 / 2025-09-26	38	3 - Scheduling
2025-09-29 / 2025-10-03	39	4 - Concurrency
2025-10-06 / 2025-10-10	40	5 - AVR UC3
2025-10-13 / 2025-10-17	41	6 - AVR FreeRTOS
2025-10-20 / 2025-10-24	42	7 - BRTT and Rpi
2025-10-27 / 2025-10-31	43	8 - BRTT and PREEMPT RT
2025-11-03 / 2025-11-07	44	Theory Exercise, in Sentralbygg 1 S3 at 18:00 (Starts 18:15)
2025-11-10 / 2025-11-14	45	
2025-11-17 / 2025-11-21	46	

I appreciate to get questions in front on mail (within Monday 2025-11-17), which make me able to prepare (hopefully good) explanations.

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Also use

Blackboard → TTK4147 Sanntidssystemer (2025 HØST) →
Learning materials → Questions

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